Supplementary Material: Measuring Coordination-Collapse Boundaries in Reward-Mixed Cooperative MARL

Anonymous submission

This document contains the apparatus supporting the main paper: full experimental setup and hyperparameters (§1), the complete loss-aversion sweep and baseline reconciliation (§2), the scoped mean-field derivation (§3), the cross-environment and cross-algorithm detail (§4), both boundary-condition ablations (§5), the pre-registered signal-to-noise probe (§6), the exploration-noise ablation (§7), threshold robustness (§8), notation (§10), and a claim–evidence ledger (§11). Section, equation, and figure references to the main paper are by name. The supply-chain collapse label, analysis code, and mechanism probes were fixed before execution; the anonymized code/data archive included with the submission contains the corresponding records and scripts.

1 Experimental Setup in Full

Primary substrate. `NetworkManagement-v1` from or-gym (Hubbs et al. 2020): a six-node multi-echelon inventory network with Poisson demand (μ=20) on the retail edge, 30-period episodes, uniform lead times. Node 1 (retail-facing) follows a fixed demand-pull policy, stabilizing the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterized via a base-stock wrapper mapping each agent's continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \mathrm{IP})$ where IP is inventory position. The substrate extends prior DRL inventory work, the Beer Distribution Game Dec-POMDP of Oroojlooyjadid et al. (2022), the multi-echelon A3C results of Gijsbrechts et al. (2022), and single-echelon MARL (Leluc et al. 2023), to a six-node network with per-node agents.

Algorithm and evaluation. The supply-chain learner is parameter-shared PPO ("IPPO" in the shared-parameter sense): a single shared actor–critic trained on the pooled per-node experience, with each node receiving its own observation and acting through its own action indices, and decentralized execution. Each trained node runs PPO (Schulman et al. 2017) on this shared-parameter MLP policy via Stable-Baselines3 2.8.0 (Raffin et al. 2021) with PettingZoo/SuperSuit vectorization (Terry et al. 2021; Terry, Black, and Hari 2020); all hyperparameters are listed in Table 1 (values not set explicitly are SB3 2.8.0 defaults, enumerated there for completeness). MAPPO uses the identical settings with a centralized (joint-state) critic. Evaluation is deterministic over 10 episodes with 20 fixed seeds per condition; the 10-episode evaluation estimates each trained policy's collapse label, while cross-run uncertainty (the bootstrap CIs) is driven by the 20 training seeds, which the seed-paired bootstrap (§9) keeps separate from evaluation noise.

Table 1: Full PPO/MAPPO hyperparameters (Stable-Baselines3 2.8.0). Only `n_steps` and `batch_size` are set explicitly; the rest are SB3 defaults, listed for reproducibility.

Hyperparameter	Value
Policy / value net	MLP, 2×64, $\tanh$ (shared params across nodes)
Centralized critic (MAPPO)	joint-state input, same 2×64 MLP
Learning rate	3×10^{-4} (constant)
Rollout length `n_steps`	128
Minibatch size	256
Update epochs	10
Discount γ	0.99
GAE λ	0.95
Clip range	0.2
Value clip	none
Entropy coef.	0.0 (except entropy ablation)
Value-loss coef.	0.5
Max grad norm	0.5
Advantage normalization	on
Parallel envs	1 (one core/run; SuperSuit concat)
Policy init	SB3 default (orthogonal, gain $\sqrt{2}$)
Optimizer	Adam

Computing infrastructure. All training is CPU-only (no GPU), one core per run on a Linux HPC cluster (PBS scheduler, `ncpus`=1; per-run walltimes of 3–12 hours scaling with the training horizon). Software versions: Python with Stable-Baselines3 2.8.0, PettingZoo 1.26.1, SuperSuit 3.11.0, Gymnasium 1.2.2, `mpe2` 1.1.0, PyTorch 2.12.0, NumPy 2.4.6, and pandas 3.0.3.

Experiment inventory. Table 2 itemizes the primary supply-chain runs, total run count 5,064, including the

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Table 2: Primary supply-chain experiment inventory (same environment and algorithm unless noted; $\alpha{=}\beta{=}1$ is the linear no-PT condition). Total 5,064 runs.

Exp	Purpose	λ	α, β	Seeds	Runs
A @150k	Struct. boundary	1	1,1	20	220
A @300k	Struct. boundary	1	1,1	20	140
C	Basin control	1	1,1	12	144
N	Null baseline	1	1,1	20	180
B-Red.	Reward-transf. decomp.	1,3,7	0.88	20	300
λ-sweep	Loss-av. law	2,4,5,6	0.88	20	400
T @450k	Metastability	1	1,1	20	160
T @900k	Metastability	1	1,1	20	160
T @600k	Metastability (curv. probe)	1	0.88	20	140
Entropy	SNR dissociation	1	1,1	20	400
High-λ grid	Steep-cell resolution	5,6,7	0.88	20	120
Ctrl A	Reference point ($4 \times z_0$)	1,3,7	0.88	20	1,200
Ctrl B	Scale match (RMS)	1,3,7	0.88	20	600
Ctrl C	Reward units (2× loc/team)	1,3,7	0.88	20	900

150k/300k structural boundaries, the 450k/900k/600k horizon points behind the metastability fit, the reward-transform decomposition and loss-aversion sweep, the finer high-λ grid, the entropy ablation, and the three reward-transform controls (§9).

Basin control (rules out initialization). Exp C rules out initialization as a collapse driver. Within-w correlations between S_{init} and converged S are -0.11 ($w{=}0.5$), -0.38 ($w{=}0.85$), and -0.36 ($w{=}1.0$), small to moderate and inconsistent with initialization dominating; the collapsed/survived assignment is insensitive to where agents start.

Code and artifact. An anonymized code bundle accompanies this submission: the environment wrappers, all experiment drivers and analysis scripts, the frozen collapse-label and control pre-registration records, and the raw per-run result CSVs sufficient to regenerate every table and figure.

2 Full Reward-Transform Sweep and Baseline Reconciliation

The recurring w_{crit} values. There is one structural null; the other recurring numbers are a horizon measurement of it and a curvature *result*, not independent baselines (Table 3).

Full loss-aversion sweep. Table 4 gives the complete $\lambda{=}1\ldots7$ sweep with Firth penalized fits. The increase is monotone up to estimation noise at the high-λ plateau: w_{crit} rises strictly from $\lambda{=}1$ to $\lambda{=}5$ then saturates near 0.84–0.85 ($\lambda{=}5, 6$ statistically indistinguishable). At high loss aversion the transition is very steep and falls between adjacent grid points (collapse fractions $1.00, 1.00, 1.00, 0.55, 0.00$ at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$ for $\lambda{=}7$), so the boundary

Table 3: The recurring w_{crit} values and the single role each plays.

w_{crit}	Condition	Role
0.666	$\alpha{=}\beta{=}1$, 300k	structural null (main sweep)
0.685	$\alpha{=}\beta{=}1$, 300k (Exp N)	linear baseline, reproduced
0.822	$\alpha{=}\beta{=}1$, 150k	null at shorter horizon
0.585	$\alpha{=}\beta{=}0.88$, $\lambda{=}1$	curvature result

Table 4: Full loss-aversion sweep ($\alpha{=}\beta{=}0.88$, 300k, 20 seeds/(w, λ) cell, $n{=}100$ runs/λ). w_{crit} with bootstrap 95% CIs; slope/p from Firth penalized logistic regression.

λ	w_{crit}	95% CI	Firth b_1	p
1	0.585	[0.545, 0.626]	-14.7	1.2×10^{-6}
2	0.715	[0.687, 0.745]	-27.2	4.4×10^{-6}
3	0.774	[0.751, 0.798]	-29.7	5.1×10^{-6}
4	0.806	[0.766, 0.841]	-36.8	5.5×10^{-6}
5	0.825	[0.806, 0.844]	-53.1	4.7×10^{-4}
6	0.835	[0.817, 0.850]	-24.7	2.5×10^{-6}
7	0.854	[0.837, 0.871]	-43.1	1.5×10^{-3}

remains identifiable (the $w{=}0.85$ cell is interior, $11/20$ collapsed) but the unpenalized MLE slope diverges. Firth penalized logistic regression is stable here; all seven slopes reject a flat relationship (max $p = 1.5 \times 10^{-3}$) and the Firth w_{crit} matches the bootstrap to within 0.019, confirming the large unpenalized slopes at $\lambda{=}5, 7$ were a fitting artifact, not separation. The high-λ plateau is a reward-transform saturation, not a $w{=}1$ ceiling artifact: at $\lambda{=}7$ no evaluated runs collapse at $w{=}0.95$ (collapse fraction 0.00), consistent with the bounded affine law in which $w_{\text{crit}}(\lambda) \to 1$ only as $\lambda \to \infty$. The $\lambda{\geq}5$ values in Table 4 use a finer $w \in \{0.80, 0.90\}$ grid added specifically to resolve the steep transition; this places two interior points across each high-λ boundary and halves the CI widths ($0.038, 0.033, 0.024$ at $\lambda{=}5, 6, 7$) relative to the single $w{=}0.85$ cell, giving the clean monotone $0.825 \to 0.835 \to 0.854$.

Metastability under matched curvature. The exploratory 600k metastability probe (Table 2, T @600k) uses $\alpha{=}\beta{=}0.88$, which shifts w_{crit} left by 0.100, so its collapse fractions are partly curvature-deflated. The main-text four-horizon slide is measured at matched $\alpha{=}\beta{=}1$ and is therefore already curvature-free; for a direct comparison at 600k under $\alpha{=}\beta{=}1$ (Exp 0b, 20 seeds), $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$, versus 0.80 and 0.40 at 300k: the horizon slide holds with curvature held constant, confirming it is not a curvature artifact.

The metastable slide generalizes. The horizon dependence is not a supply-chain artifact. We ran dedicated horizon sweeps under centralized-critic MAPPO on both transfer environments. On MPE Cooperative Navigation, w_{crit} falls $0.82 \to 0.68$ as the horizon grows $2\text{M} \to 8\text{M}$ env steps (slide $+0.14$); on Level-Based Foraging it falls $0.33 \to 0.20$ over $1\text{M} \to 4\text{M}$ (slide $+0.13$), each with non-overlapping bootstrap intervals at the endpoints. Both match the supply-chain

Table 5: Matched risk-neutral horizons (α=β=1, λ=1). w_{crit} with bootstrap 95% CIs; the four points fit $w_{\text{crit}} = 2.44 - 0.138 \ln T$ at RMSE 0.027.

T (env steps)	w_{crit}	95% CI
150k	0.822	[0.785, 0.858]
300k	0.666	[0.635, 0.694]
450k	0.622	[0.603, 0.641]
900k	0.573	[0.547, 0.598]

direction (boundary slides left with more training), so the metastability result (like the loss-aversion direction) holds across environments and algorithms, not just on the primary substrate.

The four-horizon log-linear fit. On the supply chain we measured w_{crit} at four matched risk-neutral horizons (Table 5). The four points fall on a log-linear law $w_{\text{crit}} = 2.44 - 0.138 \ln T$ (RMSE 0.027), the Kramers escape-time prediction that w_{crit} falls as $-\ln T$ toward the deterministic floor $w^0_{\text{crit}} = c/(c+g)$. All four CI widths are $\leq$0.073, and the 300k/450k/900k widths are $\leq$0.054, so the law is a fitted four-point relationship, not a two-point extrapolation.

2.1 Is collapse a distinct regime? Continuous-metric distributions

A fitted 50% crossing exists whenever the binary label varies monotonically with w, so the substantive question is whether there are genuinely distinct behavioral regimes or a continuous degradation. Figure 1 plots the continuous metric per run across w at λ=1. A point-estimate bimodality coefficient a hair above a descriptive rule-of-thumb, with no uncertainty and no formal test, is not sufficient evidence for a distinct dynamical regime; we address this directly.

Bootstrap CI and Hartigan's dip test. For each substrate and w we compute a bootstrap 95% CI on Sarle's bimodality coefficient (2000 resamples) and Hartigan's dip test against a unimodal null (`experiments/bimodality_significance.py`). On the supply chain, the mean base-stock $\bar{S}$ shows the same qualitative pattern as before (runs cluster near $\bar{S} \approx 0$ or in $[45, 320]$, e.g. $8/20$ runs at $\bar{S}$=0 at w=0.65; coefficient $0.59, 0.57$ at w=$0.65, 0.75$), but the bootstrap CI is wide and straddles 0.555 at every w (e.g. $[0.43, 0.78]$ at w=0.65, $[0.45, 0.74]$ at w=0.75), and the dip test does not reject unimodality at any w ($p \geq 0.30$, n=20/cell). On LBF, the raw team-return dip test is nominally significant at high w ($p = 0.014, 0.002, 0.002$ at w=$0.75, 0.85, 0.95$), which would seem to favor bimodality on the substrate we had called a smooth continuum. Inspecting the raw values shows this is a ceiling artifact: 7–8 of 40 runs per cell saturate exactly at the metric's upper bound. Excluding ceiling-clipped runs removes the significance at every w ($p \geq 0.11$).

Conclusion. Neither substrate clears a formal multimodality test at our sample sizes (n=20 per supply-chain cell, n=40 per LBF cell). We therefore withdraw the "genuine attractor" claim for the supply chain and describe "collapse" uniformly, on both substrates, as a threshold-defined behavioral crossing rather than a confirmed distinct regime. This also means the "metastable escape" framing (§3) should be read as describing the horizon-dependence of a threshold crossing, not basin-level escape dynamics; we do not have dwell-time, escape-time, or hysteresis evidence for either substrate. Distinguishing a true attractor from a steep continuum would need larger per-cell n, unbounded (non-ceiling-clipped) continuous metrics, and trajectory-level probes; we leave this to future work.

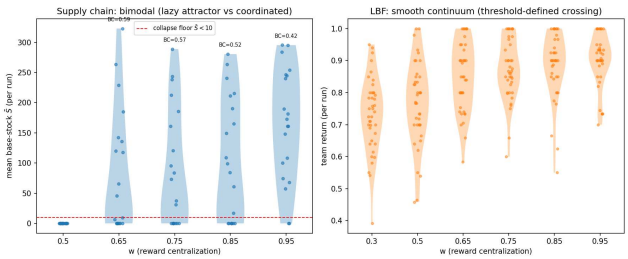


Figure 1: Per-run continuous metric across w (λ=1). Left: supply-chain mean base-stock $\bar{S}$ shows the same qualitative clustering near $\bar{S} \approx 0$ vs. coordinated, with an empty gap; the collapse floor $\bar{S} < 10$ is dashed; a dip test does not reject unimodality at any w (§2.1). Right: LBF team return, with the ceiling-clipped runs responsible for the raw dip-test signal visible as the pileup at 1.0.

The derived affine law (18) and its out-of-sample fit (RMSE 0.014 across all seven points) are shown and discussed in the main paper; the derivation of its form is given in §3 below.

2.2 Estimator robustness (link function and nonparametric fit)

We refit every w_{crit} under three estimators: logistic (primary), probit, and isotonic regression (monotone-decreasing in w, no functional form). Across all λ and all three environments the estimates agree to $\leq$ 0.02, and the isotonic w_{crit} (which assumes only monotonicity) tracks the logistic to within one grid step, so the reported location and its CI are not consequences of the logistic link. On the primary supply-chain sweep the logistic/Firth and isotonic $w_{\text{crit}}(\lambda)$ coincide at $\{0.585/0.575,\ 0.774/0.786,\ 0.854/0.859\}$ for $\lambda \in \{1, 3, 7\}$, and the λ=1$\to$7 shift is $+0.269$ (logistic/Firth) versus $+0.284$ (isotonic); McFadden pseudo-R^2 for the logistic fits is ≥ 0.6 at every interior cell. The same agreement holds on MPE and LBF, where the seed-paired λ-shift CI excludes zero under all three estimators.

3 Mean-Field Mechanism: Scope and Derivation

This section gives the mechanism underlying the sign-level claims in the main paper. Its scope is deliberately narrow. The derivation is a mean-field account for the primary supply-chain PPO setting, where agents use continuous base-stock actions and the reward-mixing dial w creates a local–global

incentive conflict. It is not a closed-form derivation of the numerical value of w_{crit}. The empirical boundary remains calibrated by the fitted collapse curve; the mechanism offers a candidate account of why a boundary exists, why its position slides with the training horizon, and why loss aversion raises it under a local signal-to-noise premise checked on that substrate. We do not have distributional or basin-level evidence for a distinct regime (§2.1), so "sharp" should be read as "well-estimated," not as evidence of a sudden dynamical transition.

3.1 Reward transform and policy-mean dynamics

Each trained node i receives the reward-mixed scalar

$$z_i(a) = (1-w)\ell_i(a_i, a_{-i}) + wG(a), \qquad G(a) = \sum_j \ell_j(a), \tag{1}$$

where ℓ_i is the local profit of node i and G is system profit. The prospect-theoretic value transform used in the experiments is

$$v(z) = \begin{cases} z^{\alpha}, & z \geq 0, \\ -\lambda(-z)^{\beta}, & z < 0. \end{cases} \tag{2}$$

Thus the shaped training signal is $r_i^{\mathrm{PT}} = v(z_i)$. We do not model probability weighting in this paper; the experimental transform consists only of value-function curvature (α, β) and loss aversion λ.

For the continuous-action PPO supply-chain substrate, let m_i denote the mean of agent i's Gaussian base-stock policy. Around a slowly varying policy state, the stochastic PPO update of m_i can be approximated by the diffusion

$$\mathrm{d}m_i = F_i(m; w, \alpha, \beta, \lambda)\,\mathrm{d}t + \sqrt{\Sigma_i(m; w, \alpha, \beta, \lambda)}\,\mathrm{d}W_i, \tag{3}$$

where F_i is the expected policy-mean drift and $\Sigma_i > 0$ is the variance of the stochastic update. We use F only for drift and Σ only for diffusion variance. Constant factors in the diffusion convention, such as whether one writes $\sqrt{\Sigma}$ or $\sqrt{2\Theta}$, are absorbed into the critical signal-to-noise threshold below and do not affect any sign claim.

3.2 The local–global incentive conflict

The primary supply-chain substrate has the incentive asymmetry required for a collapse boundary. Along the base-stock direction, increasing an upstream node's stock is locally costly because it increases holding and procurement costs, but globally beneficial because it prevents downstream stockouts. A finite-difference probe of the substrate gives the local signs

$$\partial_a \mathbb{E}[\ell_i] \approx -c, \qquad \partial_a \mathbb{E}[G] \approx +g, \qquad c, g > 0. \tag{4}$$

Ignoring the prospect transform and stochastic-update variance, the cooperative component of the policy-mean drift has the form

$$F_0(w) = \eta\{wg - (1-w)c\}. \tag{5}$$

The deterministic zero-drift crossing is therefore

$$w_{\mathrm{crit}}^0 = \frac{c}{c+g}. \tag{6}$$

This is not the empirical boundary measured in the paper. It is only the low-noise, infinite-horizon floor toward which the finite-horizon boundary can move. At finite horizon, stochastic updates and basin geometry make the operative boundary a signal-to-noise crossing, not merely the point where the deterministic drift changes sign.

3.3 Collapse as failure to escape the lazy basin

Let $x \in [0, 1]$ be a latent order parameter measuring progress from the lazy basin toward coordinated behavior. We do not claim that x is directly observed in the experiments. It is a macroscopic coordinate used to describe the basin-level dynamics:

$$\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, w, \lambda, N)}\,\mathrm{d}W_t. \tag{7}$$

Here F is the effective drift toward coordination and Σ is the stochastic-update variance. Let $\tau_{L\to C}(w, \lambda)$ denote the mean first-passage time to escape the lazy basin and reach the coordinating basin.

This convention fixes the direction of the argument. A run is collapsed at training horizon T when it has not escaped the lazy basin by the end of training:

$$\text{collapse at horizon } T \quad \Longleftrightarrow \quad \tau_{L\to C}(w, \lambda) > T \tag{8}$$

up to the usual stochastic first-passage variability. The empirical midpoint $w_{\mathrm{crit}}^{\mathrm{Emp}}(T, \lambda)$ is the value of w at which the fitted collapse probability equals $1/2$, equivalently the finite-horizon crossing where the typical escape time is on the order of T.

Increasing w strengthens the cooperative global component of the reward, so it decreases $\tau_{L\to C}$. Increasing loss aversion weakens the effective escape signal relative to update noise in the loss domain, so it increases $\tau_{L\to C}$. Therefore a larger w is required to reach the same fixed-horizon crossing when λ increases.

3.4 Signal-to-noise form of the boundary

The quantity that matters near the boundary is not drift alone, but escape signal-to-noise. Define the local escape SNR

$$\mathcal{S}(w, \lambda) = \frac{F_{\mathrm{esc}}(w, \lambda)}{\sqrt{\Sigma_{\mathrm{esc}}(w, \lambda)}}, \tag{9}$$

where F_{esc} and Σ_{esc} denote the effective drift and update variance on the escape interval from the lazy basin to the coordinating basin. The finite-horizon boundary is the crossing

$$\mathcal{S}(w, \lambda) \approx \kappa_T, \tag{10}$$

where κ_T is a horizon-dependent critical value. Longer training horizons require a smaller escape SNR, so $w_{\mathrm{crit}}(T)$ decreases with T. This gives the metastable interpretation of the observed horizon slide: at any fixed training budget the boundary is well-defined, but additional training can rescue some policies below the shorter-horizon boundary.

Equivalently, in the one-dimensional diffusion approximation, Kramers' first-passage form gives

$$\tau_{L\to C}(w, \lambda) \asymp \tau_0 \exp\{\Delta U_{L\to C}(w, \lambda)\}, \tag{11}$$

where $\Delta U_{L\to C}$ is the effective quasi-potential barrier for leaving the lazy basin. Larger escape SNR lowers this barrier and shortens $\tau_{L\to C}$; lower escape SNR raises the barrier and lengthens $\tau_{L\to C}$. The numerical value of $\Delta U_{L\to C}$ depends on the unobserved covariance structure of PPO updates, so the mechanism does not produce a closed-form numerical w_{crit}.

3.5 Sign propagation under a checked local premise

The sign claim used in the main paper is conditional on a local premise, checked on the supply-chain substrate, about how w and λ affect the escape SNR.

Premise. On the supply-chain PPO escape interval,

$$\partial_w \mathcal{S}(w,\lambda) > 0, \qquad \partial_\lambda \mathcal{S}(w,\lambda) < 0. \tag{12}$$

The first inequality states that increasing centralization strengthens the escape signal. The second states that increasing loss aversion lowers the effective escape signal-to-noise ratio. This second effect may arise through drift, diffusion, or both: loss aversion changes the transformed reward landscape in the loss domain and can increase stochastic update variance relative to the cooperative escape signal. We therefore state the premise at the SNR level rather than identifying it with drift alone.

Proposition 1 (Loss aversion raises the empirical boundary)**.** *Assume the local SNR premise above and assume the mean escape time $\tau_{L\to C}$ is strictly decreasing in $\mathcal{S}$. Then, at fixed training horizon T,*

$$\frac{\mathrm{d}w_{\mathrm{crit}}^{\mathrm{Emp}}}{\mathrm{d}\lambda} > 0. \tag{13}$$

Proof. The empirical boundary is the fixed-horizon crossing

$$\tau_{L\to C}(w,\lambda) = T. \tag{14}$$

Since $\tau_{L\to C}$ decreases with escape SNR and $\partial_w \mathcal{S} > 0$, we have $\partial_w \tau_{L\to C} < 0$. Since $\partial_\lambda \mathcal{S} < 0$, increasing loss aversion lengthens the lazy-to-coordination escape time: $\partial_\lambda \tau_{L\to C} > 0$. Applying the implicit function theorem to $H(w,\lambda) = \tau_{L\to C}(w,\lambda) - T = 0$ gives

$$\frac{\mathrm{d}w_{\mathrm{crit}}^{\mathrm{Emp}}}{\mathrm{d}\lambda} = -\frac{\partial_\lambda \tau_{L\to C}}{\partial_w \tau_{L\to C}} > 0. \tag{15}$$

The proposition proves only the direction. Its magnitude depends on the unobserved diffusion and basin geometry, which is why the empirical boundary is measured rather than predicted in closed form. □

3.6 Curvature is scoped to the primary substrate

The same SNR language explains why value-function curvature and loss aversion can move the boundary in opposite directions on the primary supply-chain PPO substrate. In that setting, concave value curvature ($\alpha, \beta < 1$) compresses reward variation and empirically raises the escape SNR relative to the linear transform, lowering w_{crit}. Loss aversion magnifies the loss-domain component and lowers the escape SNR, raising w_{crit}.

This curvature sign is not claimed to be universal. It is a primary-substrate result. The transfer experiments show that the loss-aversion direction generalizes, while the curvature sign and exact functional form are substrate-specific. Thus the mechanism supports the supply-chain decomposition and the sign of the loss-aversion shift, but it does not claim a global curvature theorem for all cooperative MARL tasks.

3.7 Affine loss-aversion law on the supply-chain substrate

The supply-chain PPO sweep is additionally summarized by a bounded one-parameter law after anchoring the baseline. At $\lambda = 1$, the curvature baseline fixes the ratio g/c through

$$w_{\mathrm{crit}}(1) = \frac{1}{1+g/c}. \tag{16}$$

Loss aversion scales only the loss-domain share of the marginal escape incentive. Writing that share as $\rho \in [0,1]$, define

$$\phi(\lambda) = 1 + \rho(\lambda - 1). \tag{17}$$

The resulting bounded form is

$$w_{\mathrm{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}. \tag{18}$$

This is not a free closed-form prediction of the numerical boundary. The baseline anchors g/c, and ρ is one additional calibrated parameter. The useful content is that the form is monotone, bounded, and saturating. Fitting ρ on $\lambda \in \{3,7\}$ and holding out $\lambda \in \{2,4,5,6\}$ gives $w_{\mathrm{crit}}^{\mathrm{pred}} = \{0.585, 0.696, 0.761, 0.802, 0.832, 0.854, 0.870\}$ against $w_{\mathrm{crit}}^{\mathrm{meas}} = \{0.585, 0.715, 0.775, 0.805, 0.820, 0.835, 0.854\}$ (Figure 2): held-out RMSE 0.014, all-seven RMSE 0.013; leave-one-out fits keep $\rho \in [0.59, 0.64]$ ($\rho = 0.63$ on the full $\lambda \in \{3,7\}$ calibration).

Two caveats bound this. First, the held-out λ share environment, algorithm, and horizon with the calibration points, so this is an interpolation check within one sweep, not out-of-distribution validation, and a weak one: held-out RMSE (0.014) is on the same scale as the held-out points' own bootstrap CI half-widths (0.017–0.027), near the measurement noise floor. We treat the form as a parsimonious summary, not a validated law. Second, the $\rho \in [0,1]$ bound is slack here (ρ=0.63) but binds on LBF, where the affine form misfits at RMSE 0.15: the loss-aversion *sign* transfers there but this specific functional form does not.

The affine law is therefore a single-substrate mechanism summary. It is not claimed to transfer to LBF or MPE, where the absolute levels, curvature sign, and functional form differ.

4 Generalization Detail

4.1 Cross-environment: MPE Cooperative Navigation

Three agents cover three landmarks in a continuous 2-D arena (discrete actions, 50-step episodes). The locally rational action under the herding incentive is to stay near teammates, collectively disastrous, as agents cluster and no landmark is covered. We set `local_ratio`=0 so the raw reward is the

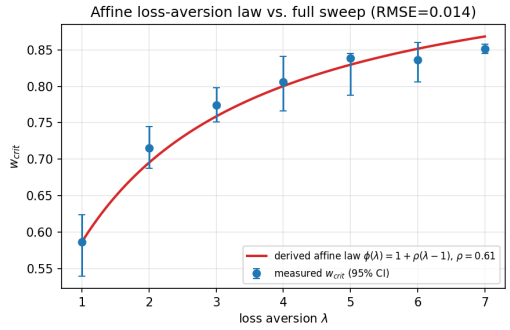


Figure 2: The one-parameter saturating summary (line, ρ=0.63 after anchoring g/c at λ=1) against measured $w_{\mathrm{crit}}(\lambda)$ (points, bootstrap 95% CIs). Held-out RMSE on $\lambda \in \{2, 4, 5, 6\}$ is 0.014; all-seven RMSE is 0.013. An interpolation check within one sweep, not out-of-distribution validation.

pure global team score; the herding term is defined by our wrapper. Collapse is per-step landmark occupancy (fraction of landmarks within a fixed radius of some agent), an evaluation metric invariant to reward scale and so unaffected by the reward transform. Unlike the supply chain (per-agent incentives), MPE's coordination reward is shared-global, which under independent learners confounds reward-transform collapse with credit-assignment failure; we therefore train MPE with centralized-critic MAPPO (Yu et al. 2022; Lowe et al. 2017), implemented as Stable-Baselines3 PPO with a centralized (joint-state) critic policy so the optimizer matches the supply-chain PPO. The herding wrapper defines each agent's local reward as

$$r_i^{\mathrm{local}} = -\,\kappa_{\mathrm{herd}} \cdot \frac{1}{N-1} \sum_{j \neq i} \|p_i - p_j\|_2, \qquad \kappa_{\mathrm{herd}} = 4, \quad (19)$$

with p_i agent i's position; the raw env uses `local_ratio=0` so r^{global} is the pure team score. The scale κ_{herd}=4 places the theoretical indifference point at $w^* \approx 0.62$ (from $(1-w)\,\overline{r_{\mathrm{herd}}} = w\,\overline{r_{\mathrm{global}}}$ with measured means 42.3 and 26.3), so the no-transform boundary is interior (w_{crit}=0.74); it was fixed before the λ-sweep. We ran the full λ-sweep (1,760 runs, 20 seeds/cell; Figure 3). The loss-aversion effect replicates, w_{crit} rises $0.78 \to 0.89$ over λ=1 $\to$ 7, rising in envelope but non-monotone at intermediate λ, excess $+0.158$ $[0.108, 0.212]$ at λ=7 and $+0.101$ $[0.055, 0.150]$ at λ=3 (matching the supply chain's $+0.089$). The curvature effect does *not* replicate ($+0.042$ $[-0.009, 0.092]$, opposite in sign), and the affine form fits MPE only to RMSE 0.05.

4.2 A third environment: Level-Based Foraging

LBF (Christianos, Schäfer, and Albrecht 2020) has a public-goods incentive: agents pay an effort cost to load food cooperatively, so the locally rational action is to free-ride. Concretely LBF is `Foraging-5x5-2p-2f-v3`: a 5×5 grid, 2 players, 2 foods, with per-agent rewards equal to each loader's share of a food (normalized so loading all foods sums to a fixed total); the 2-player 5×5 configuration is deliberately small so the w=1 team-summed reward remains learnable. Natively discrete, unrelated to logistics or navigation, trained under centralized-critic MAPPO (matching MPE so only the environment varies), full λ-sweep, same collapse-midpoint criterion: the team return (sum of per-agent shares over an evaluation episode) below θ=0.626, the calibrated collapsed-versus-coordinated midpoint (1,600 runs). The risk-neutral null is low and tight ($w_{\mathrm{crit}} \approx 0.08$); curvature moves it *right* by $+0.18$ (opposite to the supply chain). Loss aversion drives w_{crit} from 0.26 at λ=1 to 0.83 at λ=4, monotone with non-overlapping endpoint CIs (the pre-registered criterion), robust to the collapse threshold. For $\lambda \geq 5$ the boundary saturates against the centralization ceiling: the logistic w_{crit} extrapolates to $w \geq 1$ (outside the measured $w \in [0, 0.95]$ grid), so these points are *not* interior estimates and we label them exploratory in the precise sense that their w_{crit} lies beyond the range any run samples, their location is bounded below by 0.95 but not upper-bounded by the data. An added intermediate level w=0.90 tightens the mid-λ estimates and confirms the saturation, but does not bring the $\lambda \geq 5$ crossings inside the grid. The supply chain's affine law does not transfer: with γ fixed from λ=1, the best-fit slope saturates at its bound and misfits LBF by RMSE 0.15, converting the form's substrate-specificity from assertion to measured result.

Table 6: What the mean-field mechanism claims after separating modeled signs from calibrated quantities.

Category	Claim
Modeled, not fitted	On the primary supply-chain PPO substrate: increasing w raises escape SNR; increasing λ lowers escape SNR; therefore loss aversion raises w_{crit}. Finite-horizon escape gives sharpness and a leftward horizon slide.
Empirically confirmed	Resolved (non-flat, tightly-estimated) threshold crossing; horizon slide; loss-aversion direction across SC, MPE, and LBF; entropy-noise dissociation where injected noise is binding.
Calibrated/fitted	Absolute w_{crit}; diffusion scale; g/c from the baseline; one additional affine-law parameter ρ after baseline anchoring.
Not claimed	Closed-form numerical w_{crit}; universal curvature sign; transfer of the affine law beyond the supply-chain PPO substrate; direct observability of the latent order parameter.

4.3 Transfer-environment label calibration

Both transfer thresholds were fixed by the same collapsed-versus-coordinated midpoint rule used for the supply chain, and, critically, set from the λ=1 (untreated) endpoints *before* the loss-aversion sweep, so no threshold was chosen after seeing a treatment outcome. For MPE, the collapse metric is

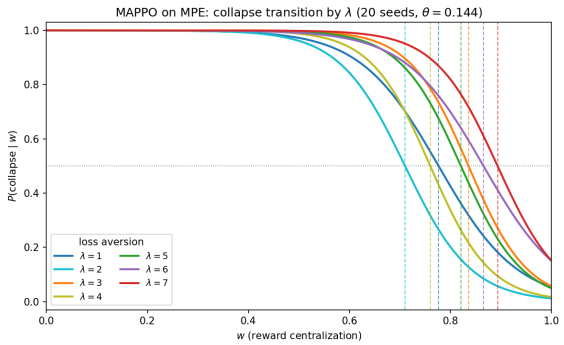


Figure 3: MAPPO on MPE Cooperative Navigation: fitted $P(\text{collapse} \mid w)$ for λ=1, . . . , 7 (α=β=0.88, centralized critic, collapse = landmark occupancy below θ=0.144). The family shifts rightward with λ, the same loss-aversion signature, under a different environment and algorithm.

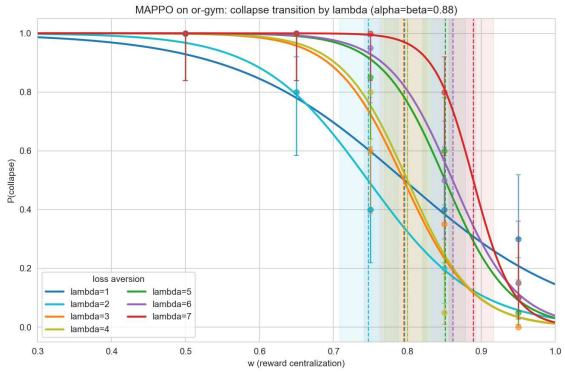


Figure 4: Centralized-critic MAPPO on the supply chain: $P(\text{collapse} \mid w)$ for λ=1, . . . , 7 (700 runs). As under PPO, the family shifts rightward with loss aversion, confirming the effect is not an artifact of independent PPO.

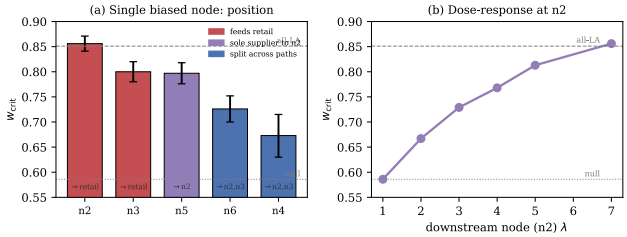


Figure 5: A single loss-averse node can dominate the supply-chain boundary. (a) One λ=7 node by position: the largest shifts come from nodes whose output reaches demand most directly, nodes feeding retail (red) dominate, the sole supplier to the dominant downstream node (purple) is comparable within noise to the second retail-feeder, and split-path suppliers (blue) are weaker as a group; CIs given in text above. (b) Dose-response: increasing the downstream node's λ moves w_{crit} monotonically up to the homogeneous all-loss-averse value (0.854, dashed).

per-step landmark occupancy; at the calibration endpoints the coordinated regime (w high) occupies ≈ 0.28 of landmarks and the free-riding regime (w=0) ≈ 0.05, and $\theta = 0.144$ is their midpoint. For LBF, the metric is episode team return; the coordinated (null, high-w) and collapsed (high-λ, low-w) regimes bracket the midpoint at $\theta = 0.626$. The occupancy and return metrics are evaluation quantities invariant to the reward transform, so the label does not move with λ. The direction result is additionally reported as robust to the threshold: on LBF the monotone λ-rise holds under $\pm$ perturbations of θ, and the pre-registered endpoint criterion (non-overlapping λ=1 vs. last-interior CIs) does not depend on the midpoint value.

4.4 Cross-algorithm: MAPPO on the supply chain

To isolate the algorithm from the environment, we re-run the full sweep on the *identical* substrate, grid, seeds, label, and hyperparameters, changing only the learner from independent PPO to centralized-critic MAPPO (700 runs; Figure 4). w_{crit} rises $0.796\,[0.742, 0.860]$ at λ=1 to $0.889\,[0.856, 0.917]$ at λ=7, monotone from λ=2 $(0.747, 0.795, 0.800, 0.851, 0.861, 0.889)$, endpoint CIs non-overlapping. The direction generalizes while the absolute level does not: MAPPO sits above PPO at low λ (most visibly λ=1, 0.796 vs 0.585, where MAPPO's transition is shallow and the estimate noisier), converging by $\lambda \geq 4$ ($|\Delta w_{\text{crit}}| \leq 0.04$).

4.5 Downstream-minority cascade (figure)

Figure 5 accompanies the minority-node analysis in the main paper (§"A Single Downstream Node Can Nearly Set the Boundary"): panel (a) is the position sweep (one λ=7 node at each network position) and panel (b) the single-downstream-node dose-response. The experiment holds the environment, PPO configuration, collapse label, w-grid, and estimator identical to the main homogeneous sweep. Bootstrap 95% CIs (Firth w_{crit}, seed resampling, 2,000 replicates, n=20 seeds/position) for panel (a): node_2 (retail-adjacent) $0.863\,[0.846, 0.880]$; node_3 (retail-adjacent) $0.800\,[0.780, 0.820]$; node_5 (sole supplier to the dominant downstream node) $0.797\,[0.740, 0.857]$; node_6 (split-path) $0.726\,[0.682, 0.769]$; node_4 (split-path) $0.673\,[0.622, 0.720]$. The coarse retail-adjacent-vs-split-path grouping is well separated; within each group the pairwise CIs overlap substantially (node_3 vs. node_5; node_4 vs. node_6), so the fine-grained ordering within a group should not be read as resolved at n=20/position.

4.6 Deconfounding position with independent actors

Because a single shared-parameter actor lets a biased node's gradient reach every other node directly, the position ordering above is not by itself evidence that the effect is topological. We designed a second sweep, identical in environment, w-grid, seeds, α, β, λ, collapse label, and estimator, but replacing the shared actor with five fully independent PPO models, one per trainable node, each updated only on its own

Table 7: Position sweep, shared vs. independent actors (λ=7, n=20/position/actor condition, 2,000-replicate seed-resampled bootstrap 95% CI). Rank 1 = strongest effect on w_{crit}.

Position	Shared w_{crit}	Rank	Independent w_{crit}	Rank
node_2	0.863 [.846, .880]	1	0.498 [.363, .581]	5
node_3	0.800 [.780, .820]	2	0.714 [.683, .745]	1
node_5	0.797 [.740, .857]	3	0.538 [.474, .591]	4
node_6	0.726 [.682, .769]	4	0.547 [.495, .594]	3
node_4	0.673 [.622, .720]	5	0.559 [.502, .608]	2

transitions and sharing no parameters or gradient.

Matching per-agent training exposure. The shared actor's nominal 300k `total_timesteps` is distributed across all 5 agents by SuperSuit's `pettingzoo_env_to_vec_env_v1`, which packs the agents into one vectorized environment; each individual agent therefore receives $\approx 300\text{k}/5 = 60\text{k}$ real environment ticks of its own transitions. To keep the two sweeps comparable, each independent actor in the separate-actor sweep is trained for 60k real ticks (round-robin in 2k-step segments across 30 rounds, each actor training against the others' current, evolving policies), matching the shared actor's real per-agent exposure rather than its nominal timestep count.

Result: the ordering is not topological. Table 7 compares the two sweeps directly. The ordering does not merely reshuffle: node_2, the strongest position under the shared actor (w_{crit}=0.863, rank 1), is the *weakest* of the five under separate actors (w_{crit}=0.498 [0.363, 0.581], rank 5), while node_4, weakest under the shared actor, rises to rank 2. The coarse retail-adjacent-vs-split-path grouping also fails to replicate: mean w_{crit} is 0.832 vs. 0.700 (a clear gap) under the shared actor, and 0.606 vs. 0.553 (no longer cleanly separated, and inverted by node_2 alone) under separate actors. We conclude the position-specific effect under the shared actor is substantially a shared-gradient artifact and do not draw a topological reading from it.

5 Two Boundary-Condition Ablations

5.1 Action representation

For the supply-chain base-stock control, the mechanism predicts that coarse discretization of the scalar action representation can remove the near-boundary crossing. Value-based learners need discrete actions, so we discretized the base-stock control into K=11 bins and ran IQL and QMIX on the identical reward pipeline, w-grid, and label; both produced a flat, w-independent collapse rate with no fittable w_{crit}. To isolate the cause, a single-variable control (PPO on the same K=11 discrete shell) also goes flat (Table 8). At K=11, discretization *alone* erases the boundary, so the value-based nulls at that granularity need no appeal to the learning paradigm. This is not a claim that discreteness per se removes the boundary: MPE is natively discrete and retains it. Coarse bins snap the policy to a stable interior level and never visit the near-threshold (s, S) region where collapse occurs, consistent with the continuous-gradient account. We confirm only the K=11 point, not the shape of the boundary's dependence on K: we cannot currently distinguish "coarseness of any granularity suppresses the boundary" from "the boundary specifically needs near-continuous control, and would reappear at finer K." A granularity sweep ($K \in \{11, 25, 51, 101\}$) on the supply-chain substrate would resolve this and is left to future work; until then we report this as a substrate-specific, single-point scope condition, not a confirmed general one.

Table 8: Action-representation ablation at K=11 (the only granularity tested). The control (last row) changes only the action space while holding the learner at PPO, isolating discretization at this granularity, not the value-based paradigm, as what erases the boundary; whether the boundary reappears at finer K is untested.

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes
QMIX (centralized value)	discrete K=11	no
IQL (independent value)	discrete K=11	no
PPO (control)	discrete K=11	**no**

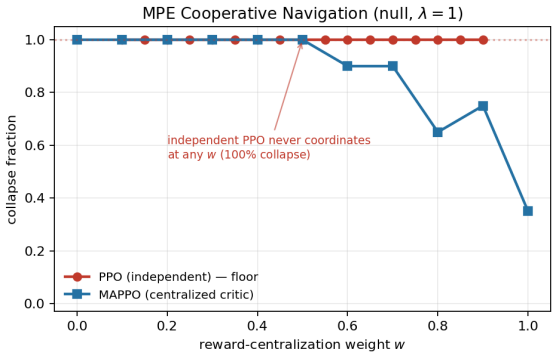


Figure 6: Credit-assignment ablation on MPE (null, λ=1). Independent PPO collapses at every w (a coordination floor), while centralized-critic MAPPO crosses the boundary. The reward-transform boundary is exposed only once credit assignment is handled.

5.2 Credit assignment

The boundary is measurable only with a learner that can coordinate on a shared-reward task. On MPE, independent PPO never reaches the coordinated regime at *any* w (collapse fraction 1.0 throughout, including maximal centralization; Figure 6), so collapse induced by the reward transform cannot be distinguished from inability to assign credit. Centralized-critic MAPPO recovers coordination (collapse fraction falls to 0.35, occupancy crosses the threshold near $w \approx 0.6$). The same floor appears on LBF. The boundary is therefore measurable on these shared-reward tasks only with a centralized critic. Matching MAPPO to such environments is a necessity, not a convenience.

6 Pre-registered Signal-to-Noise Probe

The mean-field account calibrates the magnitude of w_{crit} rather than predicting it, because the drift and diffusion coefficients are properties of the stochastic PPO learning dynamics. We pre-registered a direct measurement intended to test whether these coefficients could localize the boundary. During PPO training we logged the policy mean once per update and, over a fixed early window, estimated an empirical drift $\widehat{F}(w)$ from mean increments and an empirical diffusion variance $\widehat{\Sigma}(w)$ from increment variance. We formed

$$\widehat{\mathcal{S}}(w) = \frac{\widehat{F}(w)}{\sqrt{\widehat{\Sigma}(w)}}. \tag{20}$$

If one uses the alternative convention $\Sigma = 2\Theta$, the constant factor is absorbed into the critical threshold and does not change any crossing or sign claim. **Boundary-reproduction check:** sweep $w \in \{0.50, \ldots, 0.90\}$ at the structural condition, 20 seeds/w, 150k steps; the measured collapse-fraction crossing should reproduce the matched-horizon boundary $w_{\text{crit}} = 0.822$. **Outcome (reported in full):** the collapse-fraction of this independent sweep crosses 0.5 at $w = 0.82$, reproducing the matched-horizon boundary almost exactly, a positive robustness check on the boundary. The *drift-sign* observable, however, was *inconclusive*: the restoring drift in the contested band is weakly positive across all w with no genuine zero-crossing, so it does not localize w_{crit}. The reason is basin geometry, PPO initializes above the cooperative optimum ($S{\approx}250$ vs $S{\approx}150$), and collapses plunge through the contested band too quickly to sample, biasing the conditioned drift toward survivors.

Non-survivor-biased re-probe (also negative). We tested the natural fix directly. We re-ran the identical sweep with the policy re-initialized at the cooperative optimum ($S{\approx}150$, actor mean biased to $a_0{=}-0.4$) so that both surviving and collapsing runs are sampled inside the contested band during early training, and added a second, initialization-robust *basin-flow* localizer (net early displacement of the policy mean from the optimum) alongside the restoring-drift statistic. The re-initialization de-biased the observable (the drift is now smooth and monotone in w rather than flat and noisy) and the collapse-fraction crossing again reproduced the boundary ($w_{\text{crit}} \approx 0.83$). But neither statistic *localizes* w_{crit}: the drift and basin-flow are monotone in w without a zero-crossing, and their steepest change falls near $w{\approx}0.6$, not at the boundary. The observable therefore tracks *where the policy sits* rather than *where the boundary is*, which the initialization change does not fix. We consequently report magnitude localization as genuinely open: the sign-level account and single-substrate functional-form summary remain useful, but a magnitude-level prediction of w_{crit} remains future work. The re-probe driver and analysis are released with the code.

7 Exploration-Noise Ablation

The SNR account predicts that injected exploration noise raises w_{crit} when it is the binding SNR term. Sweeping the PPO entropy coefficient at the structural condition (300k, 20 seeds, $w \in \{0.5, \ldots, 0.9\}$), the supply-chain boundary rises monotonically: $w_{\text{crit}} = 0.685$ at ent=0 (reproducing the null), 0.690 at 0.001, 0.715 at 0.01, and beyond 0.05 the system collapses at every tested w (Table 9). **A binding-noise dissociation.** Repeating under MAPPO on the transfer substrates: on LBF the prediction replicates (w_{crit} rises $0.13 \to 0.37$ over ent $\in \{0, 0.05\}$, shift $+0.24$, non-overlapping endpoints); on MPE the boundary does *not* move (shift $+0.05$, overlapping CIs; pooled slope over 1,320 runs positive but not significant, $\beta{=}+0.47$, CI $[-3.4, +4.3]$). This is a dissociation predicted by the mechanism: exploration noise moves the boundary only when it is the binding term in the SNR. The supply chain and LBF have sparse, sharp landscapes where injected entropy dominates; MPE's dense distance-shaped reward carries high intrinsic exploration that swamps the added term. The pattern matches the binding-noise criterion: the boundary moves where injected entropy is binding and is statistically flat where it is not.

Table 9: Entropy-coefficient ablation ($\alpha{=}\beta{=}1$, $\lambda{=}1$, 300k, 20 seeds/cell). w_{crit} with bootstrap 95% CIs.

entropy coef.	w_{crit} (95% CI)
0.000	0.685 [0.656, 0.710]
0.001	0.690 [0.667, 0.715]
0.010	0.715 [0.689, 0.743]
0.050	> 1 (collapse at all w)

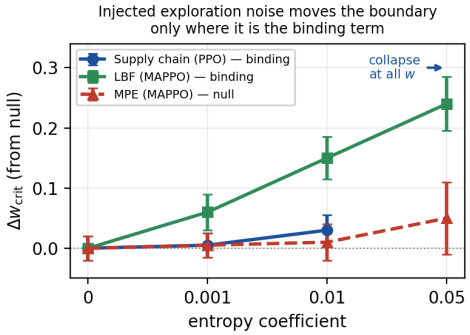


Figure 7: Entropy-ablation dissociation across substrates. Injected exploration noise raises w_{crit} clearly on LBF (non-overlapping CIs); the supply-chain shift is same-direction but within noise (overlapping CIs); MPE is flat, where dense-reward intrinsic exploration dominates.

8 Threshold Robustness

We re-estimate w_{crit} on Exp A @300k under nine variants of the collapse definition, varying the $\bar{S}$ threshold over $\{8, 10, 12\}$ and the $\bar{\pi}$ threshold over $\{-102.5, -127.1, -152.5\}$. Three variants with $\bar{\pi} \leq -152.5$ are degenerate (the minimum profit in the dataset is -128.1) and excluded. Across the remaining six estimable variants the maximum deviation from the baseline $w_{\text{crit}} = 0.666$ is 0.015, well below the CI width: the collapse definition is robust to $\pm 20\%$ threshold perturbation.

9 Reward-Transform Controls

The loss-aversion treatment also changes reward magnitude, variance, and (through the fixed zero reference) the gain/loss split. We isolate these with three matched controls on the supply chain, holding the environment, PPO configuration (§1), w-grid, seeds (42–61), and collapse label fixed; only the named factor moves. Reference points and the RMS scale were measured on λ=1 (untreated) data and frozen before the sweeps (pre-registration records released with the code).

What "same seed" means, and paired vs. unpaired uncertainty. The paired-bootstrap comparison depends on what is actually shared between two runs carrying the same integer seed s, since assigning a common label does not by itself make observations statistically paired. In our driver, s sets three things jointly: the PPO network's parameter initialization (`PPO(..., seed=s)`), the environment's reset/transition RNG stream, and the evaluation episode seeds (deterministic functions of s). So seed-s runs across w-cells and λ arms start from the same initial weights and draw from the same underlying randomness stream, not merely a shared bookkeeping label; treating s as the resampling unit (a block/cluster bootstrap) is the standard practice this justifies. The bootstrap resamples whole seeds with replacement, 2,000 replicates; every reported difference (across λ or control arm) draws the *same* resampled seed multiset for both endpoints within a replicate (seed-paired), which is what lets us separate evaluation-episode noise from training-seed noise for the difference specifically.

To check whether pairing is understating uncertainty, we also computed an *unpaired* version: the two endpoints are resampled independently, so cancellation from shared resampling noise cannot occur. For the same endpoint convention as the main-paper generalization table, the paired shifts are supply chain $+0.269$, MPE $+0.11$, and LBF $+0.57$ (using LBF's last interior endpoint, λ=4, rather than the ceiling-extrapolated higher-λ fits). The unpaired check gives the same signs and only marginally wider intervals, so pairing is not materially shrinking the reported conclusions.

Control A (reference point). The value transform is not invariant to adding a constant, so we re-run $\lambda \in \{1, 3, 7\}$ with the gain/loss reference z_0 shifted to four frozen values and re-anchor $v(x - z_0)$. *Control B (scale match).* We divide the transformed reward by a per-λ constant equal to its RMS relative to λ=1 (measured RMS $7.93, 17.06, 31.87$ at λ=1, 3, 7), matching magnitude across conditions. *Control C (reward units).* We multiply the local, or the team, reward by 2 before blending, changing the nominal-to-effective mixing ratio.

Reading: the λ=1→7 shift is positive with a zero-excluding CI in every arm of A and B and in all three arms of C, so it is not an artifact of the reference point or of reward magnitude/variance. In C the absolute w_{crit} moves ($+0.194$ at λ=1 under 2× local, -0.109 under 2× team), confirming that w is a nominal mixture coefficient whose numerical boundary is conditional on reward parameterization.

The incentive ratio ϱ, reported numerically. The main paper uses $\varrho = (1 - w_{\text{crit}})c_\ell/(w_{\text{crit}}c_g)$ as the coordinate-adjusted quantity; here we report its value directly from Control C, where c_ℓ, c_g are the pre-registered reward-unit multipliers of that arm (c_ℓ=2, c_g=1 for 2× local; c_ℓ=1, c_g=2 for 2× team; c_ℓ=c_g=1 at baseline), using the same Firth w_{crit} and seed-resampling bootstrap as elsewhere (`experiments/compute_varrho.py`, reads existing Control C data, no new training).

Table 10: Reward-transform controls (supply chain, PPO, 300k, 20 seeds). Shift is the seed-paired λ=1→7 change in w_{crit} (Firth); CI = 2,000-resample paired bootstrap. Baseline (reference 0, unit scale) shift is $+0.269\ [0.232, 0.306]$.

Control	Arm	$w_{\text{crit}}(\lambda$=1)	Shift λ=1→7
A ref. point	$z_0 = -2.99$ (mean)	0.605	$+0.228\ [0.187, 0.272]$
A ref. point	$z_0 = -6.77$ (median)	0.626	$+0.183\ [0.122, 0.269]$
A ref. point	$z_0 = -5.14\ (-0.5\sigma)$	0.621	$+0.193\ [0.141, 0.247]$
B scale match	RMS-matched	0.585	$+0.145\ [0.107, 0.185]$
C reward unit	2× local	0.779	$+0.130\ [0.105, 0.153]$
C reward unit	2× team	0.476	$+0.329\ [0.263, 0.430]$

Table 11: Incentive ratio $\varrho = (1-w_{\text{crit}})c_\ell/(w_{\text{crit}}c_g)$ by Control C arm and λ. Bootstrap 95% CI, 2,000 seed resamples.

Arm	c_ℓ, c_g	λ	w_{crit}	ϱ (95% CI)
Baseline	1, 1	1	0.585	$0.710\ [0.596, 0.836]$
Baseline	1, 1	3	0.775	$0.291\ [0.266, 0.325]$
Baseline	1, 1	7	0.854	$0.171\ [0.148, 0.195]$
2× local	2, 1	1	0.779	$0.568\ [0.500, 0.651]$
2× local	2, 1	3	0.880	$0.272\ [0.222, 0.306]$
2× local	2, 1	7	0.909	$0.200\ [0.175, 0.222]$
2× team	1, 2	1	0.476	$0.550\ [0.449, 0.798]$
2× team	1, 2	3	0.726	$0.189\ [0.167, 0.214]$
2× team	1, 2	7	0.805	$0.121\ [0.098, 0.145]$

ϱ is *not* itself invariant across arms at fixed λ: at λ=1 the three arms give $0.710, 0.568, 0.550$, a spread of 0.16 with CIs that overlap but are not coincident, so ϱ under this nominal (unit-multiplier) definition does not fully absorb the reward-unit change. It is, however, tighter than the raw boundary: at λ=1 the reward-unit spread in w_{crit} itself is 0.30 (0.780 to 0.476 across the two 2× arms), roughly twice ϱ's spread of 0.16; the compression ratio is 0.53 at λ=1, weakening to 0.66 at λ=3 and 0.76 at λ=7 (Table 11), so ϱ is a partial, not complete, correction that degrades somewhat as λ grows. What does hold, in every arm without exception, is the *direction*: ϱ falls monotonically from λ=1 to λ=7, by a comparable relative amount (-76% baseline, -65% 2× local, -78% 2× team), so the loss-aversion shift is not an artifact of the reward-unit convention even though ϱ's absolute level is convention-dependent. We considered whether an RMS-based (measured, rather than nominal multiplier) definition of c_ℓ, c_g would rescue full invariance, but the pre-blend local and global reward streams were not separately logged during Control C's runs (only the post-blend value was recorded), and no model checkpoints were retained, so testing this would require a new training sweep rather than reanalysis; we leave it to future work and report the nominal-scale result here without overstating it.

10 Notation and Abbreviations

Symbol	Meaning
w	reward centralization weight, $\in [0, 1]$
w_{crit}	critical w where $P(\mathrm{collapse})=0.5$
λ	loss-aversion coefficient ($\lambda>1$)
α, β	prospect-theoretic value-function curvature (gain/loss)
ρ	loss-domain share of marginal incentive, $\in [0, 1]$
g, c	marginal global benefit / local cost of cooperation
F, Σ	policy-mean drift / stochastic-update diffusion variance
$\mathcal{S}(w, \lambda)$	$F_{\mathrm{esc}}/\sqrt{\Sigma_{\mathrm{esc}}}$, escape signal-to-noise ratio
σ, η, B	exploration scale / learning rate / batch size
$\Delta U, \tau$	escape barrier / Kramers first-passage time
T	training horizon (environment steps)
$\bar{S}, \bar{\pi}$	mean base-stock / mean profit (collapse label)

Abbreviations. MARL (multi-agent reinforcement learning); CTDE (centralized training, decentralized execution); PT value transform (prospect-theoretic value transform); PPO (proximal policy optimization); MAPPO (multi-agent PPO with a centralized critic); MPE (multi-agent particle environments); LBF (Level-Based Foraging); SNR (signal-to-noise ratio); CI (confidence interval).

11 Claim–Evidence Ledger

Table 12 assigns each headline claim to its evidence bucket. The loss-aversion direction is the generalizing core; the curvature sign, affine law, and downstream-minority cascade are supply-chain-specific unless separately replicated.

Table 12: Claim ledger after the final scope pass. SC = supply chain.

Claim	Evidence bucket
Sharp collapse boundary w_{crit} exists	measured generally (SC, MPE, LBF; PPO/MAPPO)
Metastability: w_{crit} slides with T	measured generally (SC, MPE, LBF; PPO/MAPPO)
Loss aversion raises w_{crit}	measured generally for direction (SC, MPE, LBF; PPO/MAPPO)
Curvature lowers w_{crit}	measured only on SC PPO; not universal
Affine λ-law	calibrated/fitted on SC PPO only
Mean-field sign account	modeled/sign-level under SC PPO SNR premise
Downstream-minority cascade	measured only on SC PPO; future work for other directed-flow domains
Closed-form numerical w_{crit}	future work

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